

KUNWAR HIMITEESH SAHAJ

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Agentic AI Engineer with hands-on experience building multi-agent LLM systems, RAG pipelines, and end-to-end AI applications. Proficient in LangChain, LangGraph, FastAPI, and Neo4j. Passionate about architecting autonomous agents that reason, retrieve, and act — from financial assistants to real-time trip planners. Currently pursuing B.Tech (ECE + IoT) at IIIT Ranchi.

EDUCATION

Indian Institute of Information Technology (IIIT) Ranchi

2024 – 2028

B.Tech – Electronics & Communication Engineering | Spec: Embedded Systems & IoT | CGPA:

8.55/10

PROJECTS

FinGuard — AI-Powered Financial Assistant

[\[GitHub\]](#)

- Designed a **multi-agent LangGraph system** with StateGraph orchestration to route queries across Government Scheme Intelligence, Scam Detection, Budget Tracking, and Financial Education modules.
- Built a **GraphRAG pipeline**: dual Neo4j databases (schemes knowledge graph + user transactions) combined with ChromaDB vector search for hybrid semantic + structured retrieval, achieving **90% scheme-recommendation accuracy**.
- Implemented **hybrid scam detection**: LLM-based pattern recognition + ML models + rule-based red-flag engine; integrated Gmail API for phishing detection with CRITICAL / HIGH / MEDIUM / LOW risk classification.
- Engineered multilingual support (English, Hindi, Hinglish) with auto-language detection; natural language expense logging and real-time budget alerts via **FastAPI** async streaming backend.

Stack: FastAPI, LangChain, LangGraph, Groq (Llama 3.1), Google Gemini, Neo4j, ChromaDB, Gmail API, Python

Agentic Trip Planner

[\[GitHub\]](#) [\[Live\]](#)

- Built a **multi-agent LangGraph system** with **StateGraph** orchestration, **tools_condition** routing, and parallel tool execution across weather, places, currency, and calculator tools.
- Developed **FastAPI** backend with async message streaming; integrated **OpenWeatherMap**, **Tavily Search**, and **AlphaVantage APIs** for real-time data retrieval within agent loops.
- Deployed as a **live Streamlit application** supporting multi-turn conversational trip planning with stateful agent memory across sessions.

Stack: LangGraph, LangChain, FastAPI, Streamlit, OpenWeatherMap API, Tavily, AlphaVantage, Python

YouTube Transcript Chatbot — Chrome Extension

- Built a **Chrome Extension** with FastAPI backend that extracts live YouTube transcripts and enables intelligent Q&A via a **RAG pipeline** (LangChain + OpenAI API).
- Implemented real-time transcript retrieval and chunked vector indexing for low-latency, context-grounded responses directly inside the browser.

Stack: FastAPI, LangChain, OpenAI API, HTML/CSS/JavaScript, Chrome Extension APIs

DDPM — Denoising Diffusion Probabilistic Models (Scratch to Pipeline)

- Implemented the full **forward and reverse diffusion pipeline** from scratch in NumPy + PyTorch; derived closed-form noise scheduling and posterior mean/variance formulas for step-by-step denoising.
- Trained a **UNet2DModel** end-to-end on the Smithsonian Butterflies dataset using HuggingFace Diffusers + DDPM Scheduler with AdamW optimizer over 30 epochs, generating novel butterfly images.
- Visualized noise schedules, training loss convergence (raw + log scale), and full denoising progression across 100–200 timesteps on synthetic and MNIST data.

Stack: PyTorch, NumPy, HuggingFace Diffusers & Datasets, Matplotlib, Jupyter

Movie Recommendation System

- Built a personalized recommender using **vector embeddings** and cosine similarity; performed EDA and feature engineering on large movie datasets for improved accuracy.

Stack: Python, scikit-learn, Pandas, NumPy

TECHNICAL SKILLS

Languages: Python, C++, C, JavaScript, HTML/CSS

Agentic AI: LangChain, LangGraph, StateGraph, Tool-Use, Multi-Agent Orchestration, RAG, GraphRAG

AI / ML: Deep Learning, CNNs, Diffusion Models, GANs, NLP, Computer Vision, Generative AI

Frameworks: FastAPI, PyTorch, TensorFlow, Keras, HuggingFace Diffusers, Streamlit, MoviePy, OpenCV

Databases: Neo4j (Graph DB), ChromaDB (Vector DB), FAISS, Pinecone, MongoDB

Tools: Git, GitHub, Jupyter, Colab, VS Code, FFmpeg

APIs: Groq, OpenAI, Google Gemini, Tavily, ElevenLabs, OpenWeatherMap, AlphaVantage, Gmail API, Replicate

RESPONSIBILITIES & LEADERSHIP

- AI/ML Wing Coordinator — House of Geeks, IIIT Ranchi**
- Aug 2024 – Present**
- Led AI/ML initiatives; mentored peers; conducted workshops on LLMs, Agentic AI, transformer architectures, and RAG pipelines for the college tech community.

EXTRACURRICULAR

- Diploma in Guitar — Prayag Sangeet Samiti, Allahabad; passionate about music and creative expression.